

LEARNING OBJECTIVES

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STUDENT TIP

Wherever your career takes you, one of the most useful tools you can have, as a manager, is the ability to manage a project.

VIDEO 3.1

Project Management at Hard Rock's Rockfest

The Importance of Project Management

When Bechtel, the subject of the opening *Global Company Profile*, begins a project, it quickly has to mobilize substantial resources, often consisting of manual workers, construction professionals, cooks, medical personnel, and even security forces. Its project management team develops a supply chain to access materials to build everything from ports to bridges, dams, and monorails. Bechtel is just one example of a firm that faces modern phenomena: growing project complexity and collapsing product/service life cycles. This change stems from awareness of the strategic value of time-based competition and a quality mandate for continuous improvement. Each new product/service introduction is a unique event—a project. In addition, projects are a common part of our everyday life. We may be planning a wedding or a surprise birthday party, remodeling a house, or preparing a semester-long class project.

Scheduling projects can be a difficult challenge for operations managers. The stakes in project management are high. Cost overruns and unnecessary delays occur due to poor scheduling and poor controls.

Projects that take months or years to complete are usually developed outside the normal production system. Project organizations within the firm may be set up to handle such jobs and are often disbanded when the project is complete. On other occasions, managers find projects just a part of their job. The management of projects involves three phases (see Figure 3.1):

1. *Planning*: This phase includes goal setting, defining the project, and team organization.
2. *Scheduling*: This phase relates people, money, and supplies to specific activities and relates activities to each other.
3. *Controlling*: Here the firm monitors resources, costs, quality, and budgets. It also revises or changes plans and shifts resources to meet time and cost demands.

We begin this chapter with a brief overview of these functions. Three popular techniques to allow managers to plan, schedule, and control—Gantt charts, PERT, and CPM—are also described.

Project Planning

Projects can be defined as a series of related tasks directed toward a major output. In some firms a **project organization** is developed to make sure existing programs continue to run smoothly on a day-to-day basis while new projects are successfully completed.

For companies with multiple large projects, such as a construction firm, a project organization is an effective way of assigning the people and physical resources needed. It is a temporary organization structure designed to achieve results by using specialists from throughout the firm.

The project organization may be most helpful when:

1. Work tasks can be defined with a specific goal and deadline.
2. The job is unique or somewhat unfamiliar to the existing organization.
3. The work contains complex interrelated tasks requiring specialized skills.

Project organization

An organization formed to ensure that programs (projects) receive the proper management and attention.

Planning the Project (Before project)

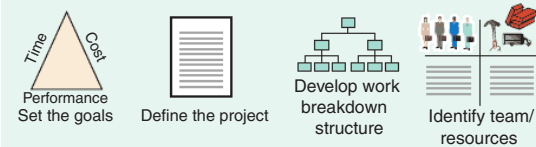
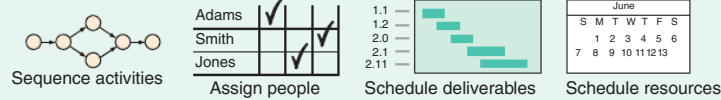


Figure 3.1

Project Planning, Scheduling, and Controlling

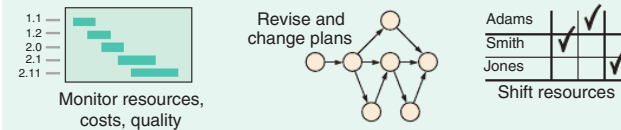
Scheduling the Project



STUDENT TIP

Managers must "make the plan and then work the plan."

Controlling the Project (During project)



4. The project is temporary but critical to the organization.
5. The project cuts across organizational lines.

The Project Manager

An example of a project organization is shown in Figure 3.2. Project team members are temporarily assigned to a project and report to the project manager. The manager heading the project coordinates activities with other departments and reports directly to top management. Project managers receive high visibility in a firm and are responsible for making sure that (1) all necessary activities are finished in proper sequence and on time; (2) the project comes in within budget; (3) the project meets its quality goals; and (4) the people assigned to the project receive the motivation, direction, and information needed to do their jobs. This means that project managers should be good coaches and communicators, and be able to organize activities from a variety of disciplines.

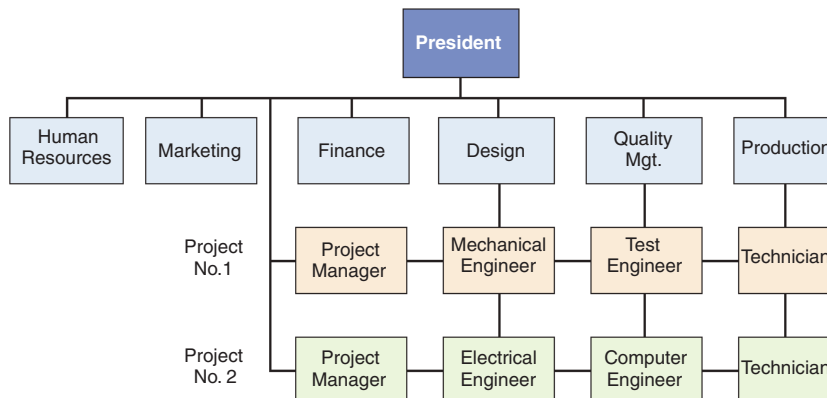


Figure 3.2

A Sample Project Organization

STUDENT TIP

Project organizations can be temporary or permanent. A permanent organization is usually called a *matrix organization*.

Ethical Issues Faced in Project Management Project managers not only have high visibility but they also face ethical decisions on a daily basis. How they act establishes the code of conduct for the project. Project managers often deal with (1) offers of gifts from contractors, (2) pressure to alter status reports to mask the reality of delays, (3) false reports for charges of time and expenses, and (4) pressures to compromise quality to meet bonuses or avoid penalties related to schedules.

Using the Project Management Institute's (www.pmi.org) ethical codes is one means of trying to establish standards. These codes need to be accompanied by good leadership and a strong organizational culture, with its ingrained ethical standards and values.

Work Breakdown Structure

The project management team begins its task well in advance of project execution so that a plan can be developed. One of its first steps is to carefully establish the project's objectives, then break the project down into manageable parts. This **work breakdown structure (WBS)** defines the project by dividing it into its major subcomponents (or tasks), which are then subdivided into more detailed components, and finally into a set of activities and their related costs. The division of the project into smaller and smaller tasks can be difficult, but is critical to managing the project and to scheduling success. Gross requirements for people, supplies, and equipment are also estimated in this planning phase.

The work breakdown structure typically decreases in size from top to bottom and is indented like this:

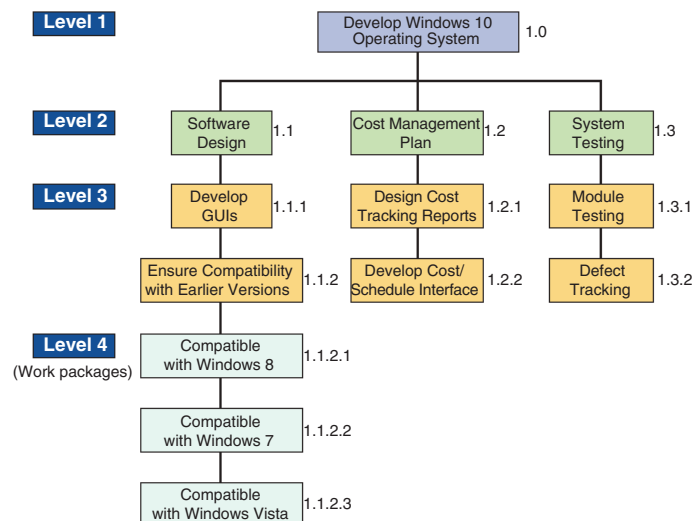
Level

1. Project
2. Major tasks in the project
3. Subtasks in major tasks
4. Activities (or "work packages") to be completed

This hierarchical framework can be illustrated with the development of Microsoft's operating system Windows 10. As we see in Figure 3.3, the project, creating a new operating system, is labeled 1.0. The first step is to identify the major tasks in the project (level 2). Three examples would be software design (1.1), cost management plan (1.2), and system testing (1.3). Two major subtasks for 1.1 are development of graphical user interfaces (GUIs) (1.1.1) and creating compatibility with previous versions of Windows (1.1.2). The major subtasks for 1.1.2 are level-4 activities, such as creating a team to handle compatibility with Windows 8 (1.1.2.1),

Figure 3.3

Work Breakdown Structure



creating a team for Windows 7 (1.1.2.2), and creating a team for Windows Vista (1.1.2.3). There are usually many level-4 activities.

Project Scheduling

Project scheduling involves sequencing and allotting time to all project activities. At this stage, managers decide how long each activity will take and compute the resources needed at each stage of production. Managers may also chart separate schedules for personnel needs by type of skill (management, engineering, or pouring concrete, for example) and material needs.

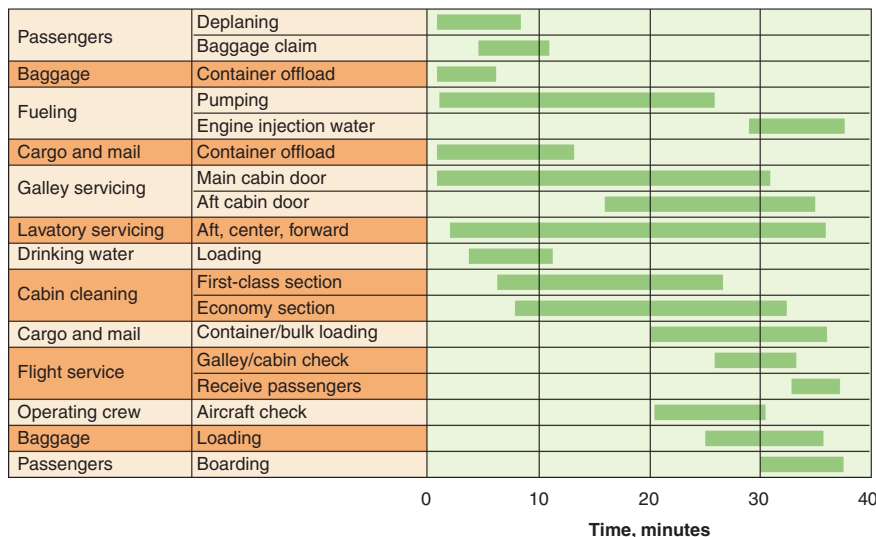
One popular project scheduling approach is the Gantt chart. **Gantt charts** are low-cost means of helping managers make sure that (1) activities are planned, (2) order of performance is documented, (3) activity time estimates are recorded, and (4) overall project time is developed. As Figure 3.4 shows, Gantt charts are easy to understand. Horizontal bars are drawn for each project activity along a time line. This illustration of a routine servicing of a Delta jetliner during a 40-minute layover shows that Gantt charts also can be used for scheduling repetitive operations. In this case, the chart helps point out potential delays. The *OM in Action* box on Delta provides additional insights.

On simple projects, scheduling charts such as these permit managers to observe the progress of each activity and to spot and tackle problem areas. Gantt charts, though, do not adequately illustrate the interrelationships between the activities and the resources.

PERT and CPM, the two widely used network techniques that we shall discuss shortly, do have the ability to consider precedence relationships and interdependency of activities. On complex projects, the scheduling of which is almost always computerized, PERT and CPM thus have an edge over the simpler Gantt charts. Even on huge projects, though, Gantt charts can be used as summaries of project status and may complement the other network approaches.

To summarize, whatever the approach taken by a project manager, project scheduling serves several purposes:

1. It shows the relationship of each activity to others and to the whole project.
2. It identifies the precedence relationships among activities.
3. It encourages the setting of realistic time and cost estimates for each activity.
4. It helps make better use of people, money, and material resources by identifying critical bottlenecks in the project.



Gantt charts

Planning charts used to schedule resources and allocate time.

STUDENT TIP

Gantt charts are simple and visual, making them widely used.

LO 3.1 Use a Gantt chart for scheduling

Figure 3.4

Gantt Chart of Service Activities for a Delta Jet during a 40-Minute Layover

Delta saves \$50 million a year with this turnaround time, which is a reduction from its traditional 60-minute routine.

OM in Action

Delta's Ground Crew Orchestrates a Smooth Takeoff

Flight 574's engines screech its arrival as the jet lumbers down Richmond's taxiway with 140 passengers arriving from Atlanta. In 40 minutes, the plane is to be airborne again.

However, before this jet can depart, there is business to attend to: passengers, luggage, and cargo to unload and load; thousands of gallons of jet fuel and countless drinks to restock; cabin and restrooms to clean; toilet holding tanks to drain; and engines, wings, and landing gear to inspect.

The 10-person ground crew knows that a miscue anywhere—a broken cargo loader, lost baggage, misdirected passengers—can mean a late departure and trigger a chain reaction of headaches from Richmond to Atlanta to every destination of a connecting flight.

Carla Sutera, the operations manager for Delta's Richmond International Airport, views the turnaround operation like a pit boss awaiting a race car. Trained crews are in place for Flight 574 with baggage carts

and tractors, hydraulic cargo loaders, a truck to load food and drinks, another to lift the cleanup crew, another to put fuel on, and a fourth to take water off. The "pit crew" usually performs so smoothly that most passengers never suspect the proportions of the effort. Gantt charts, such as the one in Figure 3.4, aid Delta and other airlines with the staffing and scheduling that are needed for this task.

Sources: *Washington Post* (September 13, 2017); *Knight Ridder Tribune Business News* (July 16, 2005); and *New York Times* (December 8, 2015).



Jeff Topping/Getty Images News/Getty Images

Project Controlling

VIDEO 3.2

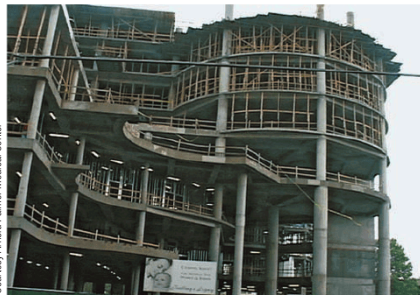
Project Management at Arnold Palmer Hospital

STUDENT TIP

To use project management software, you first need to understand the next two sections in this chapter.

The control of projects, like the control of any management system, involves close monitoring of resources, costs, quality, and budgets. Control also means using a feedback loop to revise the project plan and having the ability to shift resources to where they are needed most. Computerized PERT/CPM reports and charts are widely available today from scores of competing software firms. Some of the more popular of these programs are Oracle Primavera (by Oracle), MindView (by Match Ware), HP Project (by Hewlett-Packard), Fast Track (by AEC Software), and Microsoft Project (by Microsoft), which we illustrate in this chapter.

These programs produce a broad variety of reports, including (1) detailed cost breakdowns, (2) labor requirements, (3) cost and hour summaries, (4) raw material and expenditure forecasts, (5) variance reports, (6) time analysis reports, and (7) work status reports.



Courtesy Arnold Palmer Medical Center



Courtesy Arnold Palmer Medical Center

Construction of the new 11-story building at Arnold Palmer Hospital in Orlando, Florida, was an enormous project for the hospital administration. The photo on the left shows the first six floors under construction. The photo on the right shows the building as completed two years later. Prior to beginning actual construction, regulatory and funding issues added, as they do with most projects, substantial time to the overall project. Cities have zoning and parking issues; the EPA has drainage and waste issues; and regulatory authorities have their own requirements, as do issuers of bonds. The \$100 million, 4-year project at Arnold Palmer Hospital is discussed in the Video Case Study at the end of this chapter.

Controlling projects can be difficult. The stakes are high; cost overruns and unnecessary delays can occur due to poor planning, scheduling, and controls. Some projects are “well-defined,” whereas others may be “ill-defined.” Projects typically only become well-defined after detailed extensive initial planning and careful definition of required inputs, resources, processes, and outputs. Well-established projects where constraints are known (e.g., buildings and roads) and engineered products (e.g., airplanes and cars) with well-defined specifications and drawings may fall into this category. Well-defined projects are assumed to have changes small enough to be managed without substantially revising plans. They use what is called a **waterfall** approach, where the project progresses smoothly, in a step-by-step manner, through each phase to completion.

But many projects, such as software development (e.g., 3-D games) and new technology (e.g., landing the Mars land rover) are ill-defined. These projects require what is known as an **agile** style of management with collaboration and constant feedback to adjust to the many unknowns of the evolving technology and project specifications. In agile projects, the project manager creates a general plan, based on overall requirements and a broad perspective of the solution. From that moment, the end project is tackled iteratively and incrementally, with each increment building on the steps preceding it. This approach emphasizes small activities (or, in the case of software projects, small chunks of code) to meet limited objectives. These activities are then delivered to end users quickly so problems can be identified and corrected. There are numerous checkpoints and feedback loops to track progress.

Most projects fall somewhere between waterfall and agile.

Waterfall projects

Projects that progress smoothly in a step-by-step manner until completed.

Agile projects

Ill-defined projects requiring collaboration and constant feedback to adjust to project unknowns.

Project Management Techniques: PERT and CPM

Program evaluation and review technique (PERT) and the **critical path method (CPM)** were both developed in the 1950s to help managers schedule, monitor, and control large and complex projects. CPM arrived first, as a tool developed to assist in the building and maintenance of chemical plants at DuPont. Independently, PERT was developed in 1958 for the U.S. Navy.

Program evaluation and review technique (PERT)

A project management technique that employs three time estimates for each activity.

Critical path method (CPM)

A project management technique that uses only one time factor per activity.

The Framework of PERT and CPM

PERT and CPM both follow six basic steps:

1. Define the project and prepare the work breakdown structure.
2. Develop the relationships among the activities. Decide which activities must precede and which must follow others.
3. Draw the network connecting all the activities.
4. Assign time and/or cost estimates to each activity.
5. Compute the *longest* time path through the network. This is called the **critical path**.
6. Use the network to help plan, schedule, monitor, and control the project.

Critical path

The computed *longest* time path(s) through a network.

Step 5, finding the critical path, is a major part of controlling a project. The activities on the critical path represent tasks that will delay the entire project if they are not completed on time. Managers can gain the flexibility needed to complete critical tasks by identifying noncritical activities and replanning, rescheduling, and reallocating labor and financial resources.

Although PERT and CPM differ to some extent in terminology and in the construction of the network, their objectives are the same. Furthermore, the analysis used in both techniques is very similar. The major difference is that PERT employs three time estimates for each activity. These time estimates are used to compute expected values and standard deviations for the activity. CPM makes the assumption that activity times are known with certainty and hence requires only one time factor for each activity.

For purposes of illustration, the rest of this section concentrates on a discussion of PERT. Most of the comments and procedures described, however, apply just as well to CPM.

PERT and CPM are important because they can help answer questions such as the following about projects with thousands of activities:

1. When will the entire project be completed?
2. What are the critical activities or tasks in the project—that is, which activities will delay the entire project if they are late?
3. Which are the noncritical activities—the ones that can run late without delaying the whole project's completion?
4. What is the probability that the project will be completed by a specific date?
5. At any particular date, is the project on schedule, behind schedule, or ahead of schedule?
6. On any given date, is the money spent equal to, less than, or greater than the budgeted amount?
7. Are there enough resources available to finish the project on time?
8. If the project is to be finished in a shorter amount of time, what is the best way to accomplish this goal at the least cost?

Network Diagrams and Approaches

The first step in a PERT or CPM network is to divide the entire project into significant activities in accordance with the work breakdown structure. There are two approaches for drawing a project network: **activity on node (AON)** and **activity on arrow (AOA)**. Under the AON convention, *nodes* designate activities. Under AOA, *arrows* represent activities. Activities consume time and resources. The basic difference between AON and AOA is that the nodes in an AON diagram represent activities. In an AOA network, the nodes represent the starting and finishing times of an activity and are also called *events*. So nodes in AOA consume neither time nor resources.

Although both AON and AOA are popular in practice, many of the project management software packages, including Microsoft Project, use AON networks. For this reason, although we illustrate both types of networks in the next examples, we focus on AON networks in subsequent discussions in this chapter.

Activity-on-node (AON)

A network diagram in which nodes designate activities.

Activity-on-arrow (AOA)

A network diagram in which arrows designate activities.

Example 1

PREDECESSOR RELATIONSHIPS FOR POLLUTION CONTROL AT MILWAUKEE PAPER

Milwaukee Paper Manufacturing had long delayed the expense of installing advanced computerized air pollution control equipment in its facility. But when the board of directors adopted a new proactive policy on sustainability, it did not just *authorize* the budget for the state-of-the-art equipment. It directed the plant manager, Julie Ann Williams, to complete the installation in time for a major announcement of the policy, on Earth Day, exactly 16 weeks away! Under strict deadline from her bosses, Williams needs to be sure that installation of the filtering system progresses smoothly and on time.

Given the following information, develop a table showing activity precedence relationships.

APPROACH► Milwaukee Paper has identified the eight activities that need to be performed for the project to be completed. When the project begins, two activities can be simultaneously started: building the internal components for the device (activity A) and the modifications necessary for the floor and roof (activity B). The construction of the collection stack (activity C) can begin when the internal components are completed. Pouring the concrete floor and installation of the frame (activity D) can be started as soon as the internal components are completed and the roof and floor have been modified.

After the collection stack has been constructed, two activities can begin: building the high-temperature burner (activity E) and installing the pollution control system (activity F). The air pollution device can be installed (activity G) after the concrete floor has been poured, the frame has been installed, and the high-temperature burner has been built. Finally, after the control system and pollution device have been installed, the system can be inspected and tested (activity H).

SOLUTION ► Activities and precedence relationships may seem rather confusing when they are presented in this descriptive form. It is therefore convenient to list all the activity information in a table, as shown in Table 3.1. We see in the table that activity A is listed as an *immediate predecessor* of activity C. Likewise, both activities D and E must be performed prior to starting activity G.

TABLE 3.1 Milwaukee Paper Manufacturing's Activities and Predecessors

ACTIVITY	DESCRIPTION	IMMEDIATE PREDECESSORS
A	Build internal components	—
B	Modify roof and floor	—
C	Construct collection stack	A
D	Pour concrete and install frame	A, B
E	Build high-temperature burner	C
F	Install pollution control system	C
G	Install air pollution device	D, E
H	Inspect and test	F, G

INSIGHT ► To complete a network, all predecessors must be clearly defined.

LEARNING EXERCISE ► What is the impact on this sequence of activities if Environmental Protection Agency (EPA) approval is required after *Inspect and Test*? [Answer: The immediate predecessor for the new activity would be H, *Inspect and Test*, with *EPA approval* as the last activity.]

Activity-on-Arrow Example

Note that in Example 1, we only list the *immediate* predecessors for each activity. For instance, in Table 3.1, because activity A precedes activity C, and activity C precedes activity E, the fact that activity A precedes activity E is *implicit*. This relationship need not be explicitly shown in the activity precedence relationships.

When there are many activities in a project with fairly complicated precedence relationships, it is difficult for an individual to comprehend the complexity of the project from just the tabular information. In such cases, a visual representation of the project, using a *project network*, is convenient and useful. A project network is a diagram of all the activities and the precedence relationships that exist between these activities in a project. Example 2 illustrates how to construct an AON project network for Milwaukee Paper Manufacturing.

It is convenient to have the project network start and finish with a unique node. In the Milwaukee Paper example, it turns out that a unique activity, H, is the last activity in the project. We therefore automatically have a unique ending node.

In situations in which a project has multiple ending activities, we include a “dummy” ending activity. We illustrate this type of situation in Solved Problem 3.1 at the end of this chapter.

LO 3.2 Draw AOA and AON networks

Example 2

AON GRAPH FOR MILWAUKEE PAPER

Draw the AON network for Milwaukee Paper, using the data in Example 1.

APPROACH In the AON approach, we denote each activity by a node. The lines, or arrows, represent the precedence relationships between the activities.

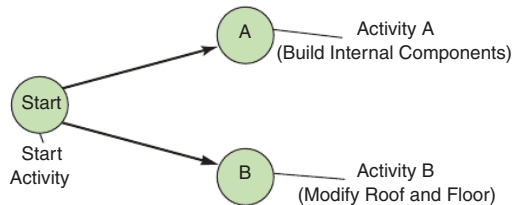
SOLUTION In this example, there are two activities (A and B) that do not have any predecessors. We draw separate nodes for each of these activities, as shown in Figure 3.5. Although not required, it is usually convenient to have a unique starting activity for a project. We have therefore included a *dummy activity* called *Start* in Figure 3.5. This *dummy activity* does not really exist and takes up zero time and resources. Activity *Start* is an immediate predecessor for both activities A and B, and it serves as the unique starting activity for the entire project.

Dummy activity

An activity having no time that is inserted into a network to maintain the logic of the network.

Figure 3.5

Beginning AON Network for Milwaukee Paper

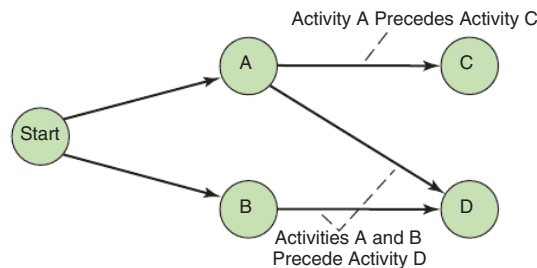


We now show the precedence relationships using lines with arrow symbols. For example, an arrow from activity *Start* to activity *A* indicates that *Start* is a predecessor for activity *A*. In a similar fashion, we draw an arrow from *Start* to *B*.

Next, we add a new node for activity *C*. Because activity *A* precedes activity *C*, we draw an arrow from node *A* to node *C* (see Figure 3.6). Likewise, we first draw a node to represent activity *D*. Then, because activities *A* and *B* both precede activity *D*, we draw arrows from *A* to *D* and from *B* to *D* (see Figure 3.6).

Figure 3.6

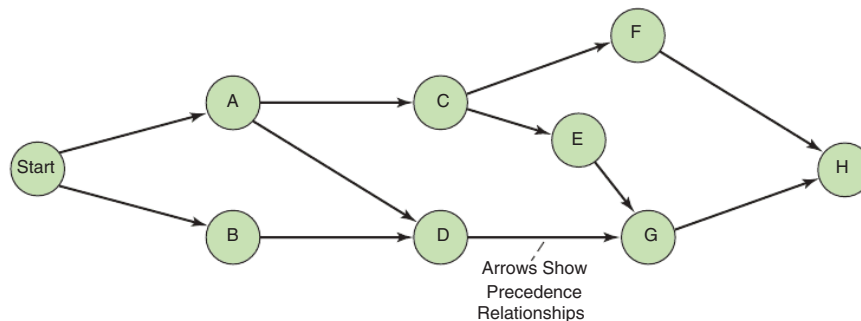
Intermediate AON Network for Milwaukee Paper



We proceed in this fashion, adding a separate node for each activity and a separate line for each precedence relationship that exists. The complete AON project network for the Milwaukee Paper Manufacturing project is shown in Figure 3.7.

Figure 3.7

Complete AON Network for Milwaukee Paper



INSIGHT Drawing a project network properly takes some time and experience. We would like the lines to be straight and arrows to move to the right when possible.

LEARNING EXERCISE If *EPA Approval* occurs after *Inspect and Test*, what is the impact on the graph? [Answer: A straight line is extended to the right beyond *H* (with a node *I* added) to reflect the additional activity.]

RELATED PROBLEMS 3.4a, 3.5, 3.8, 3.9, 3.10, 3.12, 3.13 (3.14 is available in [MyLab Operations Management](#))

Activity-on-Arrow Example

In an AOA project network we can represent activities by arrows. A node represents an *event*, which marks the start or completion time of an activity. We usually identify an event (node) by a number.

Example 3

ACTIVITY-ON-ARROW FOR MILWAUKEE PAPER

Draw the complete AOA project network for Milwaukee Paper's problem.

APPROACH▶ Using the data from Table 3.1 in Example 1, draw one activity at a time, starting with A.

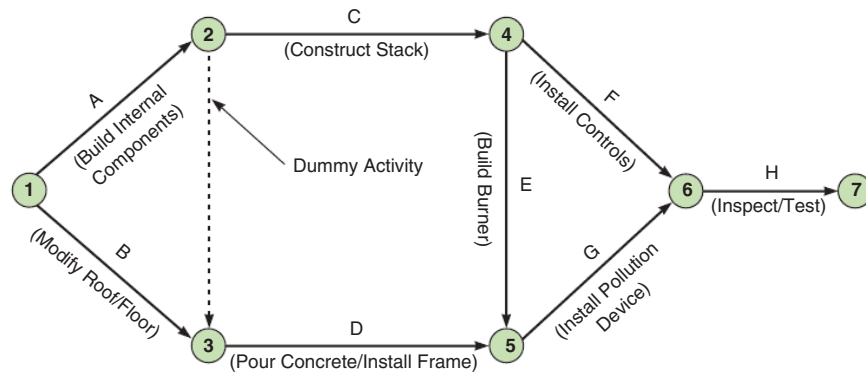
SOLUTION▶ We see that activity A starts at event 1 and ends at event 2. Likewise, activity B starts at event 1 and ends at event 3. Activity C, whose only immediate predecessor is activity A, starts at node 2 and ends at node 4. Activity D, however, has two predecessors (i.e., A and B). Hence, we need both activities A and B to end at event 3, so that activity D can start at that event. However, we cannot have multiple activities with common starting and ending nodes in an AOA network. To overcome this difficulty, in such cases, we may need to add a dummy line (activity) to enforce the precedence relationship. The dummy activity, shown in Figure 3.8 as a dashed line, is inserted between events 2 and 3 to make the diagram reflect the precedence between A and D. The remainder of the AOA project network for Milwaukee Paper's example is also shown.

Figure 3.8

Complete AOA Network
(with Dummy Activity)
for Milwaukee Paper

STUDENT TIP

The dummy activity consumes no time, but note how it changes precedence. Now activity D cannot begin until *both* B and the dummy are complete.



INSIGHT▶ Dummy activities are common in AOA networks. They do not really exist in the project and take zero time.

LEARNING EXERCISE▶ A new activity, *EPA Approval*, follows activity H. Add it to Figure 3.8. [Answer: Insert an arrowed line from node 7, which ends at a new node 8, and is labeled I (EPA Approval).]

RELATED PROBLEMS▶ 3.4b, 3.6, 3.7, 3.11a

Determining the Project Schedule

Look back at Figure 3.7 (in Example 2) for a moment to see Milwaukee Paper's completed AON project network. Once this project network has been drawn to show all the activities and their precedence relationships, the next step is to determine the project schedule. That is, we need to identify the planned starting and ending time for each activity.

Let us assume Milwaukee Paper estimates the time required for each activity, in weeks, as shown in Table 3.2. The table indicates that the total time for all eight of the company's activities is 25 weeks. However, because several activities can take place simultaneously, it is clear that the total project completion time may be less than 25 weeks. To find out just how long the project will take, we perform the **critical path analysis** for the network.

Critical path analysis

A process that helps determine a project schedule.

TABLE 3.2 Time Estimates for Milwaukee Paper Manufacturing

ACTIVITY	DESCRIPTION	TIME (WEEKS)
A	Build internal components	2
B	Modify roof and floor	3
C	Construct collection stack	2
D	Pour concrete and install frame	4
E	Build high-temperature burner	4
F	Install pollution control system	3
G	Install air pollution device	5
H	Inspect and test	2
	Total time (weeks)	25

STUDENT TIP ⚠

Does this mean the project will take 25 weeks to complete? No. Don't forget that several of the activities are being performed at the same time. It would take 25 weeks if they were done sequentially.

As mentioned earlier, the critical path is the *longest* time path through the network. To find the critical path, we calculate two distinct starting and ending times for each activity. These are defined as follows:

Earliest start (ES) = earliest time at which an activity can start, assuming all predecessors have been completed

Earliest finish (EF) = earliest time at which an activity can be finished

Latest start (LS) = latest time at which an activity can start so as to not delay the completion time of the entire project

Latest finish (LF) = latest time by which an activity has to finish so as to not delay the completion time of the entire project

We use a two-pass process, consisting of a forward pass and a backward pass, to determine these time schedules for each activity. The early start and finish times (ES and EF) are determined during the **forward pass**. The late start and finish times (LS and LF) are determined during the backward pass.

Forward pass

A process that identifies all the early times.

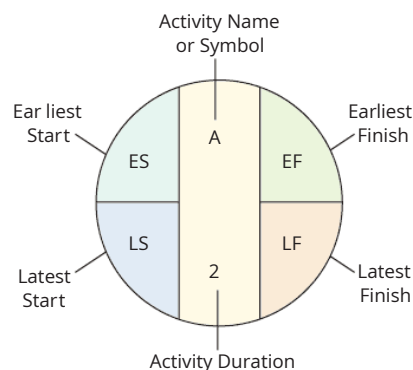
LO 3.3 Complete forward and backward passes for a project

Forward Pass

To clearly show the activity schedules on the project network, we use the notation shown in Figure 3.9. The ES of an activity is shown in the top left corner of the node denoting that activity. The EF is shown in the top right corner. The latest times, LS and LF, are shown in the bottom-left and bottom-right corners, respectively.

Figure 3.9

Notation Used in Nodes for Forward and Backward Pass



Earliest Start Time Rule Before an activity can start, *all* its immediate predecessors must be finished:

- ◆ If an activity has only a single immediate predecessor, its ES equals the EF of the predecessor.
- ◆ If an activity has multiple immediate predecessors, its ES is the maximum of all EF values of its predecessors. That is:

$$ES = \text{Max} \{EF \text{ of all immediate predecessors}\} \quad (3-1)$$

STUDENT TIP

All predecessor activities must be completed before an activity can begin.

Earliest Finish Time Rule The earliest finish time (EF) of an activity is the sum of its earliest start time (ES) and its activity time. That is:

$$EF = ES + \text{Activity time} \quad (3-2)$$

Example 4

COMPUTING EARLIEST START AND FINISH TIMES FOR MILWAUKEE PAPER

Calculate the earliest start and finish times for the activities in the Milwaukee Paper Manufacturing project.

APPROACH► Use Table 3.2, which contains the activity times. Complete the project network for the company's project, along with the ES and EF values for all activities.

SOLUTION► With the help of Figure 3.10, we describe how these values are calculated.

Because activity Start has no predecessors, we begin by setting its ES to 0. That is, activity Start can begin at time 0, which is the same as the beginning of week 1. If activity Start has an ES of 0, its EF is also 0, since its activity time is 0.

Next, we consider activities A and B, both of which have only Start as an immediate predecessor. Using the earliest start time rule, the ES for both activities A and B equals zero, which is the EF of activity Start. Now, using the earliest finish time rule, the EF for A is 2 ($= 0 + 2$), and the EF for B is 3 ($= 0 + 3$).

Since activity A precedes activity C, the ES of C equals the EF of A ($= 2$). The EF of C is therefore 4 ($= 2 + 2$).

We now come to activity D. Both activities A and B are immediate predecessors for D. Whereas A has an EF of 2, activity B has an EF of 3. Using the earliest start time rule, we compute the ES of activity D as follows:

$$ES \text{ of } D = \text{Max}\{EF \text{ of } A, EF \text{ of } B\} = \text{Max}(2, 3) = 3$$

The EF of D equals 7 ($= 3 + 4$). Next, both activities E and F have activity C as their only immediate predecessor. Therefore, the ES for both E and F equals 4 ($= EF \text{ of } C$). The EF of E is 8 ($= 4 + 4$), and the EF of F is 7 ($= 4 + 3$).

Activity G has both activities D and E as predecessors. Using the earliest start time rule, its ES is therefore the maximum of the EF of D and the EF of E. Hence, the ES of activity G equals 8 ($= \text{maximum of } 7 \text{ and } 8$), and its EF equals 13 ($= 8 + 5$).

Finally, we come to activity H. Because it also has two predecessors, F and G, the ES of H is the maximum EF of these two activities. That is, the ES of H equals 13 ($= \text{maximum of } 13 \text{ and } 7$). This implies that the EF of H is 15 ($= 13 + 2$). Because H is the last activity in the project, this also implies that the earliest time in which the entire project can be completed is 15 weeks.

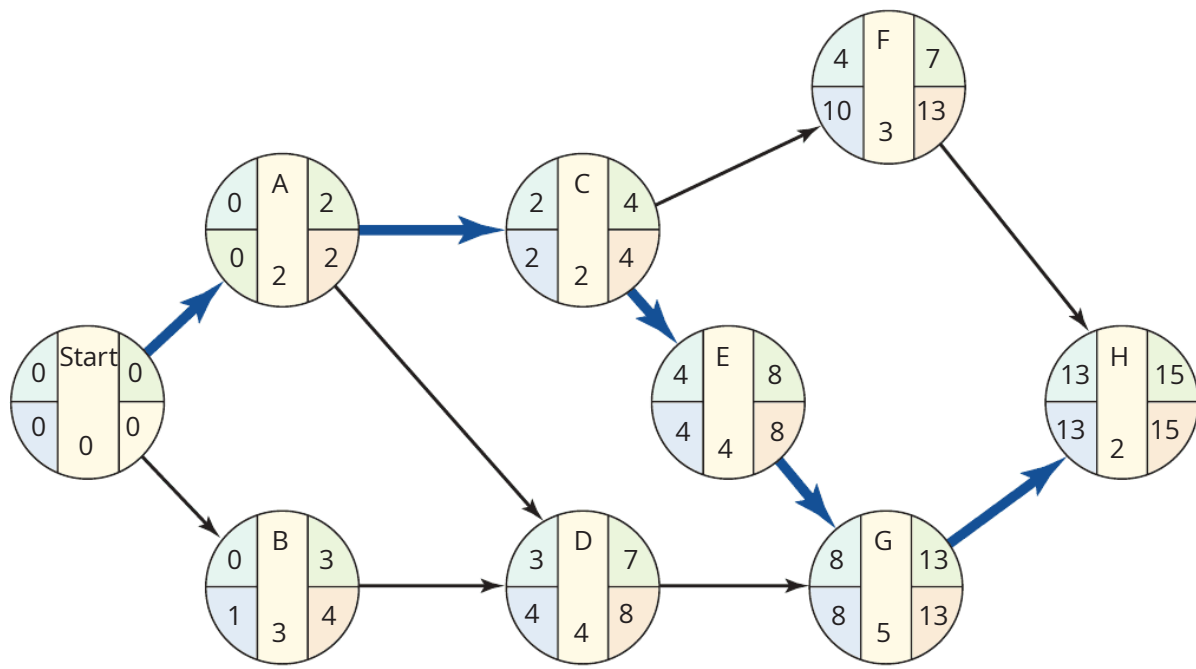
INSIGHT► The ES of an activity that has only one predecessor is simply the EF of that predecessor. For an activity with more than one predecessor, we must carefully examine the EFs of all immediate predecessors and choose the largest one.

LEARNING EXERCISE► A new activity I, *EPA Approval*, takes 1 week. Its predecessor is activity H. What are I's ES and EF? [Answer: 15, 16]

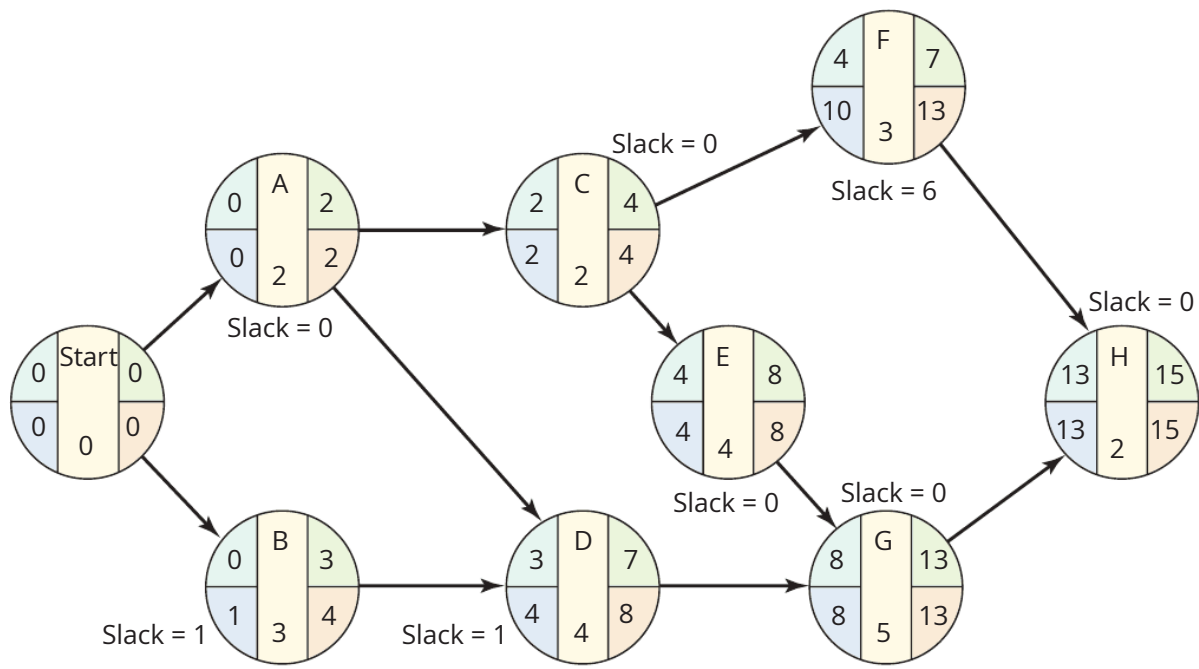
RELATED PROBLEMS► 3.15, 3.19c, 3.26c (3.16 is available in [MyLab Operations Management](#))

EXCEL OM Data File Ch03Ex4.xls can be found in [MyLab Operations Management](#).

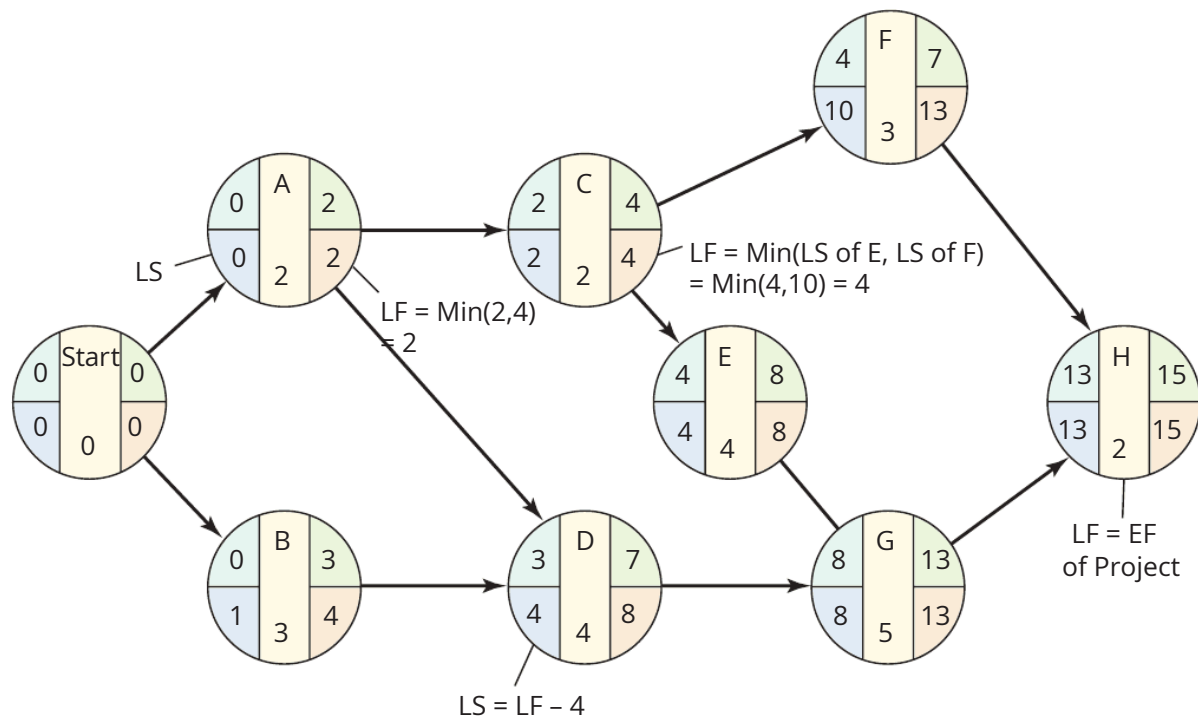
Although the forward pass allows us to determine the earliest project completion time, it does not identify the critical path. To identify this path, we need to now conduct the backward pass to determine the LS and LF values for all activities.



Overlay 3: The Critical Path Is Now Shown in Five Thick Blue Lines.



Overlay 2: Slack Times Are Now Computed and Added.



Overlay 1: Latest Start and Finish Times Are Now Added.

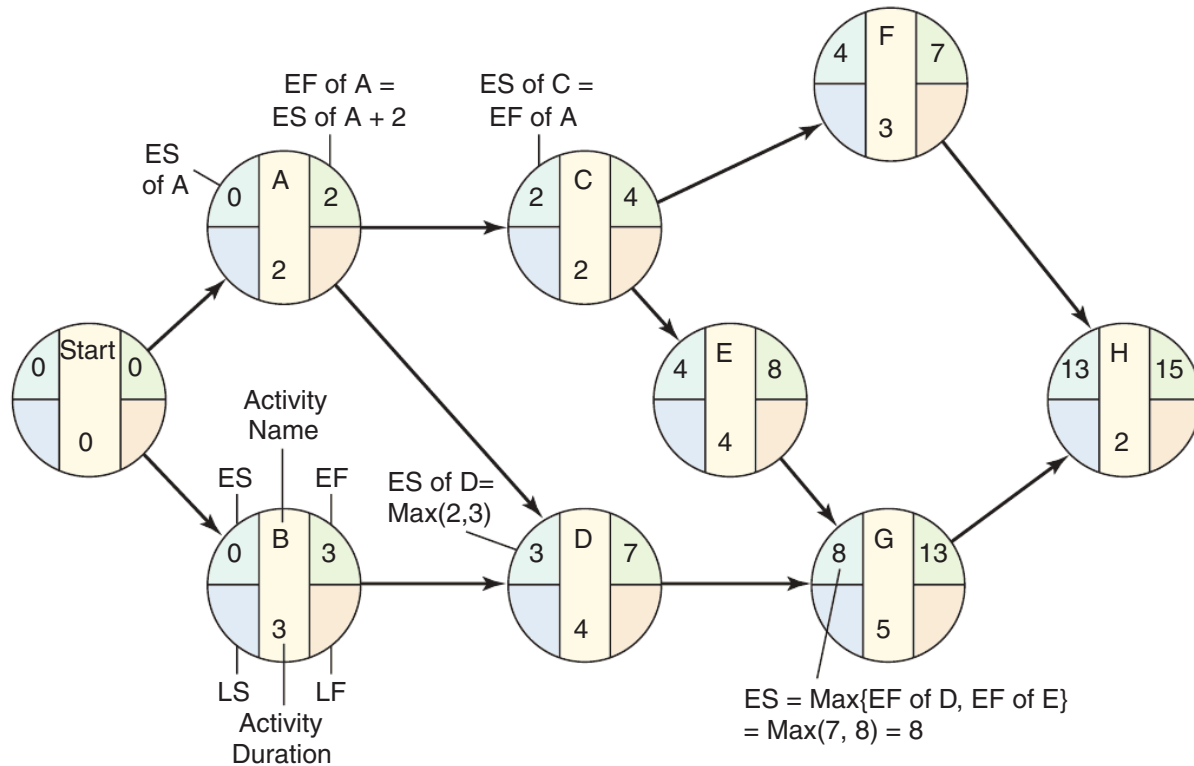


Figure 3.10 Earliest Start and Earliest Finish Times for Milwaukee Paper

Backward Pass

Backward pass

An activity that finds all the late start and late finish times.

Just as the forward pass began with the first activity in the project, the **backward pass** begins with the last activity in the project. For each activity, we first determine its LF value, followed by its LS value. The following two rules are used in this process.

Latest Finish Time Rule This rule is again based on the fact that before an activity can start, all its immediate predecessors must be finished:

- ◆ If an activity is an immediate predecessor for just a single activity, its LF equals the LS of the activity that immediately follows it.
- ◆ If an activity is an immediate predecessor to more than one activity, its LF is the minimum of all LS values of all activities that immediately follow it. That is:

$$LF = \text{Min}\{\text{LS of all immediate following activities}\} \quad (3-3)$$

Latest Start Time Rule The latest start time (LS) of an activity is the difference of its latest finish time (LF) and its activity time. That is:

$$LS = LF - \text{Activity time} \quad (3-4)$$

Example 5

COMPUTING LATEST START AND FINISH TIMES FOR MILWAUKEE PAPER

Calculate the latest start and finish times for each activity in Milwaukee Paper's pollution project.

APPROACH ► Use Figure 3.10 as a beginning point. Overlay 1 of Figure 3.10 shows the complete project network for Milwaukee Paper, along with added LS and LF values for all activities. In what follows, we see how these values were calculated.

SOLUTION► We begin by assigning an LF value of 15 weeks for activity H. That is, we specify that the latest finish time for the entire project is the same as its earliest finish time. Using the latest start time rule, the LS of activity H is equal to 13 ($= 15 - 2$).

Because activity H is the lone succeeding activity for both activities F and G, the LF for both F and G equals 13. This implies that the LS of G is 8 ($= 13 - 5$), and the LS of F is 10 ($= 13 - 3$).

Proceeding in this fashion, we see that the LF of E is 8 ($=$ LS of G), and its LS is 4 ($= 8 - 4$). Likewise, the LF of D is 8 ($=$ LS of G), and its LS is 4 ($= 8 - 4$).

We now consider activity C, which is an immediate predecessor to two activities: E and F. Using the latest finish time rule, we compute the LF of activity C as follows:

$$\text{LF of C} = \text{Min}\{\text{LS of E, LS of F}\} = \text{Min}(4, 10) = 4$$

The LS of C is computed as 2 ($= 4 - 2$). Next, we compute the LF of B as 4 ($=$ LS of D) and its LS as 1 ($= 4 - 3$).

We now consider activity A. We compute its LF as 2 ($=$ minimum of LS of C and LS of D). Hence, the LS of activity A is 0 ($= 2 - 2$). Finally, both the LF and LS of activity Start are equal to 0.

INSIGHT► The LF of an activity that is the predecessor of only one activity is just the LS of that following activity. If the activity is the predecessor to more than one activity, its LF is the smallest LS value of all activities that follow immediately.

LEARNING EXERCISE► A new activity I, *EPA Approval*, takes 1 week. Its predecessor is activity H. What are I's LS and LF? [Answer: 15, 16]

RELATED PROBLEMS► 3.15, 3.19c (3.26c is available in [MyLab Operations Management](#))

Calculating Slack Time and Identifying the Critical Path(s)

After we have computed the earliest and latest times for all activities, it is a simple matter to find the amount of **slack time** that each activity has. Slack is the length of time an activity can be delayed without delaying the entire project. Mathematically:

$$\text{Slack} = \text{LS} - \text{ES} \quad \text{or} \quad \text{Slack} = \text{LF} - \text{EF} \quad (3-5)$$

Slack time

Free time for an activity. Also referred to as free float or free slack.

Example 6

CALCULATING SLACK TIMES FOR MILWAUKEE PAPER

Calculate the slack for the activities in the Milwaukee Paper project.

APPROACH► Start with the data in Overlay 1 of Figure 3.10 in Example 5 and develop Table 3.3 one line at a time.

SOLUTION► Table 3.3 summarizes the ES, EF, LS, LF, and slack time for all of the firm's activities. Activity B, for example, has 1 week of slack time because its LS is 1 and its ES is 0 (alternatively, its LF is 4 and its EF is 3). This means that activity B can be delayed by up to 1 week, and the whole project can still be finished in 15 weeks.

On the other hand, activities A, C, E, G, and H have *no* slack time. This means that none of them can be delayed without delaying the entire project. Conversely, if plant manager Julie Ann Williams wants to reduce the total project times, she will have to reduce the length of one of these activities.

Overlay 2 of Figure 3.10 shows the slack computed for each activity.

INSIGHT► Slack may be computed from either early/late starts or early/late finishes. The key is to find which activities have zero slack.

TABLE 3.3 Milwaukee Paper's Schedule and Slack Times

ACTIVITY	ACTIVITY TIME	EARLIEST START ES	EARLIEST FINISH EF	LATEST START LS	LATEST FINISH LF	SLACK LS – ES	ON CRITICAL PATH
A	2	0	2	0	2	0	Yes
B	3	0	3	1	4	1	No
C	2	2	4	2	4	0	Yes
D	4	3	7	4	8	1	No
E	4	4	8	4	8	0	Yes
F	3	4	7	10	13	6	No
G	5	8	13	8	13	0	Yes
H	2	13	15	13	15	0	Yes

LEARNING EXERCISE ▶ A new activity I, *EPA Approval*, follows activity H and takes 1 week. Is it on the critical path? [Answer: Yes, it's $LS - ES = 0$]

RELATED PROBLEMS ▶ 3.8d, 3.11, 3.15d, 3.19c

ACTIVE MODEL 3.1 This example is further illustrated in Active Model 3.1 in [MyLab Operations Management](#).

LO 3.4 Determine a critical path

The activities with zero slack are called *critical activities* and are said to be on the critical path. The critical path is a continuous path through the project network that:

- ◆ Starts at the first activity in the project (Start in our example).
- ◆ Terminates at the last activity in the project (H in our example).
- ◆ Includes only critical activities (i.e., activities with no slack time).

Example 7

SHOWING CRITICAL PATH WITH BLUE ARROWS

Show Milwaukee Paper's critical path and find the project completion time.

APPROACH ▶ We use Table 3.3 and Overlay 3 of Figure 3.10. Overlay 3 of Figure 3.10 indicates that the total project completion time of 15 weeks corresponds to the longest path in the network. That path is Start-A-C-E-G-H in network form. It is shown with thick blue arrows.

INSIGHT ▶ The critical path follows the activities with slack = 0. This is considered the longest path through the network.

LEARNING EXERCISE ▶ Why are activities B, D, and F not on the path with the thick blue line? [Answer: They are not critical and have slack values of 1, 1, and 6 weeks, respectively.]

RELATED PROBLEMS ▶ 3.5–3.11, 3.19b, 3.21a (3.16 is available in [MyLab Operations Management](#))

Total Slack Time Look again at the project network in Overlay 3 of Figure 3.10. Consider activities B and D, which have slack of 1 week each. Does it mean that we can delay *each* activity by 1 week, and still complete the project in 15 weeks? The answer is no.

Let's assume that activity B is delayed by 1 week. It has used up its slack of 1 week and now has an EF of 4. This implies that activity D now has an ES of 4 and an EF of 8. Note that these are also its LS and LF values, respectively. That is, activity D also has no slack time now. Essentially, the slack of 1 week that activities B and D had is, for that path, *shared* between them. Delaying either activity by 1 week causes not only that activity, but also the other activity, to lose its slack. This type of a slack time is referred to as *total slack*. Typically, when two or more noncritical activities appear successively in a path, they share total slack.



Christina Kennedy/Alamy Stock Photo

To plan, monitor, and control the huge number of details involved in sponsoring a rock festival attended by more than 100,000 fans, managers use Microsoft Project and the tools discussed in this chapter. The *Video Case Study* "Managing Hard Rock's Rockfest," at the end of the chapter, provides more details of the management task.

Variability in Activity Times

In identifying all earliest and latest times so far, and the associated critical path(s), we have adopted the CPM approach of assuming that all activity times are known and fixed constants. That is, there is no variability in activity times. However, in practice, it is likely that activity completion times vary depending on various factors.

For example, building internal components (activity A) for Milwaukee Paper Manufacturing is estimated to finish in 2 weeks. Clearly, supply-chain issues such as late arrival of materials, absence of key personnel, and so on could delay this activity. Suppose activity A actually ends up taking 3 weeks. Because A is on the critical path, the entire project will now be delayed by 1 week to 16 weeks. If we had anticipated completion of this project in 15 weeks, we would obviously miss our Earth Day deadline.

Although some activities may be relatively less prone to delays, others could be extremely susceptible to delays. For example, activity B (modify roof and floor) could be heavily dependent on weather conditions. A spell of bad weather could significantly affect its completion time.

This means that we cannot ignore the impact of variability in activity times when deciding the schedule for a project. PERT addresses this issue.

STUDENT TIP

PERT's ability to handle three time estimates for each activity enables us to compute the probability that we can complete the project by a target date.

Three Time Estimates in PERT

In PERT, we employ a probability distribution based on three time estimates for each activity, as follows:

Optimistic time (a) = time an activity will take if everything goes as planned. In estimating this value, there should be only a small probability (say, 1/100) that the activity time will be $< a$.

Pessimistic time (b) = time an activity will take assuming very unfavorable conditions. In estimating this value, there should also be only a small probability (also 1/100) that the activity time will be $> b$.

Most likely time (m) = most realistic estimate of the time required to complete an activity.

Optimistic time

The "best" activity completion time that could be obtained in a PERT network.

Pessimistic time

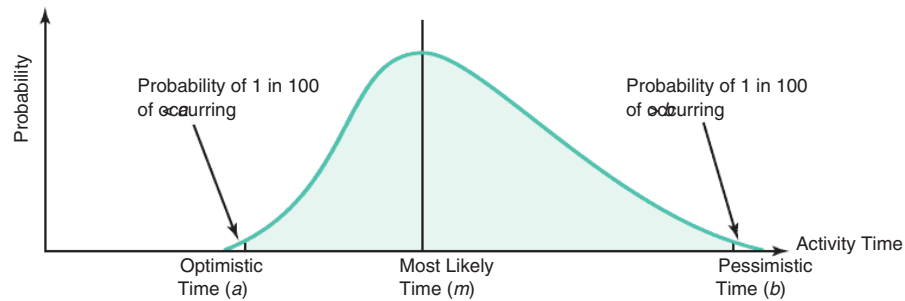
The "worst" activity time that could be expected in a PERT network.

Most likely time

The most probable time to complete an activity in a PERT network.

When using PERT, we often assume that activity time estimates follow the beta probability distribution (see Figure 3.11). This continuous distribution is often appropriate for determining the expected value and variance for activity completion times.

Figure 3.11

Beta Probability Distribution
with Three Time Estimates

To find the *expected activity time*, t , the beta distribution weights the three time estimates as follows:

$$t = (a + 4m + b)/6 \quad (3-6)$$

That is, the most likely time (m) is given four times the weight as the optimistic time (a) and pessimistic time (b). The time estimate t computed using Equation (3-6) for each activity is used in the project network to compute all earliest and latest times.

To compute the *dispersion* or *variance of activity completion time*, we use the formula:¹

$$\text{Variance} = [(b - a)/6]^2 \quad (3-7)$$

Example 8

EXPECTED TIMES AND VARIANCES FOR MILWAUKEE PAPER

Julie Ann Williams and the project management team at Milwaukee Paper want an expected time and variance for Activity F (Installing the Pollution Control System) where:

$$a = 1 \text{ week}, m = 2 \text{ weeks}, b = 9 \text{ weeks}$$

APPROACH► Use Equations (3-6) and (3-7) to compute the expected time and variance for F.

SOLUTION► The expected time for Activity F is:

$$t = \frac{a + 4m + b}{6} = \frac{1 + 4(2) + 9}{6} = \frac{18}{6} = 3 \text{ weeks}$$

The variance for Activity F is:

$$\text{Variance} = \left[\frac{(b - a)}{6} \right]^2 = \left[\frac{(9 - 1)}{6} \right]^2 = \left(\frac{8}{6} \right)^2 = \frac{64}{36} = 1.78$$

INSIGHT► Williams now has information that allows her to understand and manage Activity F. The expected time is, in fact, the activity time used in our earlier computation and identification of the critical path.

LEARNING EXERCISE► Review the expected times and variances for all of the other activities in the project. These are shown in Table 3.4.

TABLE 3.4 Time Estimates (in weeks) for Milwaukee Paper's Project

ACTIVITY	OPTIMISTIC a	MOST LIKELY m	PESSIMISTIC b	EXPECTED TIME $t = (a + 4m + b)/6$	VARIANCE $[(b - a)/6]^2$
A	1	2	3	2	$[(3 - 1)/6]^2 = 4/36 = .11$
B	2	3	4	3	$[(4 - 2)/6]^2 = 4/36 = .11$
C	1	2	3	2	$[(3 - 1)/6]^2 = 4/36 = .11$
D	2	4	6	4	$[(6 - 2)/6]^2 = 16/36 = .44$
E	1	4	7	4	$[(7 - 1)/6]^2 = 36/36 = 1.00$
F	1	2	9	3	$[(9 - 1)/6]^2 = 64/36 = 1.78$
G	3	4	11	5	$[(11 - 3)/6]^2 = 64/36 = 1.78$
H	1	2	3	2	$[(3 - 1)/6]^2 = 4/36 = .11$

STUDENT TIP

Can you see why the variance is higher in some activities than in others? Note the spread between the optimistic and pessimistic times.

RELATED PROBLEMS► 3.17a, b, 3.18, 3.19a, 3.20a (3.26b, 3.27 are available in [MyLab Operations Management](#))

EXCEL OM Data File Ch03Ex8.xls can be found in [MyLab Operations Management](#).